11-Apr-24

To

M/s. PIONEER GROUP,
DHAKA,
BANGLADESH,

Dear Sir,

Our Ref: Techno Commercial Proposal for ZLD based Filtration Plant with BRINEX System.
[Proposal No:26.Mar.2024/WTT-0736/BAN/NP-FP&BRX/6M/R0/TCP].

Dear Sir,

WTT International Private Limited (WTT) is pleased to provide M/s. PIONEER GROUP with the offer for supply of engineering & materials for ZLD based Filtration Plant with BRINEX System of capacity 6000 M^3/Day. In developing this offer WTT worked with you in an effort to understand your project and business needs. The attached proposal outlines the solutions we feel will best meet these objectives.

We greatly appreciate your consideration of WTT for this project. Our measure of success is how well we deliver solutions that help our client to meet their critical business objectives. We hope to have the opportunity to demonstrate this with your good selves.

Yours Sincerely,

D. Venkatesh
Managing Director

Proposal for:

1) ZLD based Filtration Plant with BRINEX System of capacity 6000 M^3/Day.

Proposal No: 26.Mar.2024/WTT-0736/BAN/NP-FP&BRX/6M/R0/TCP

S.NO	DESCRIPTION	DATE	PREPARED	APPROVED
1.	Techno Commercial Proposal for – "ZLD based Filtration Plant with BRINEX System".	11-Apr-24	BA - GET AR - AGM	DV - MD

CONFIDENTIALITY

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WE FEED DELIGHTED TO PRESENT OUR PATENTED SYSTEMS

Award For Excellence In Water Management

CII (Confederation of Indian Industry) *awarded*
WTT International *with the prestigious*

"Excellence in Water Management Award"

recognizing the advanced water treatment technologies
innovated by ***WTT International***
As a part of Industrial Water / Waste Management competition 2023

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1. Technical And Engineering Details

1.1. Basis of Design

- ✓ *Submerged Ceramic MBR system proposed to reduce Fouling Potential of RO feed water*
- ✓ *Contaminants removal system proposed to remove slowly Bio-degradable compounds present in the Effluent water*
- ✓ *Degasser tower proposed to reduce alkalinity/bicarbonate by stripping CO2 from effluent*
- ✓ *Reverse Osmosis System proposed to achieve recovery of **93% in 4 Stages** with **Single Skid***
- ✓ ***BRINEX System** proposed to increase the water recovery by incorporating **Hardness & Silica** removal with Reject RO Systems*
- ✓ *Reject RO system proposed to achieve recovery of **60% having a Reject TDS of about 129191 mg/L**, thereby reducing the reject water flow rate & load to the evaporator*

The following sections summarize the treatment processes, the major system components, the main system design parameters and design outputs as well as design details of system components. This section does not define scope of supply. Scope delineation is covered under section 3.

1.2. Influent Quality

S.NO	PARAMETER	UNIT	FILTRATION PLANT INLET
1.	pH	--	6.5 - 7.5
2.	BOD	mg/L	< 30
3.	COD	mg/L	< 100
4.	TDS	mg/L	4000
5.	TSS	mg/L	30

6.	TKN ^	mg/L	< 1
7.	P_4	mg/L	< 2
8.	PVA	mg/L	NIL
9.	SULPHATE	mg/L	700
10.	CHLORIDE	mg/L	600
11.	REACTIVE SILICA #	mg/L	< 15
12.	HARDNESS *	mg/L as $CaCO_3$	200
13.	ALKALINITY	mg/L as $CaCO_3$	400
14.	TEMPERATURE	ºC	30 – 35
15.	OIL & GREASE ^	mg/L	NIL
16.	IRON ^	mg/L	< 0.01

IMPORTANT

NOTE: *Since the *Parametrical values are not available (or) mismatching from the ETP inlet to MBR inlet values, it has been considered for the purpose of designing the treatment system.*

The above-mentioned ^ values are the recommended concentrations to be allowed at the Filtration System inlet. The presence of TKN, Oil & Grease and Iron will significantly affect the life and performance of the membrane in the long run and requires proper pre-treatment in existing ETP.

REACTIVE SILICA #The above-mentioned values are the maximum concentrations to be allowed at the Filtration inlet which should not be exceeded and may try avoiding the usage of Sodium silicate in the process plant. (Instead of using raw water in textile process, may reuse the RO treated water because here the reactive Silica at our RO Permeate water would be less than 1 mg/L).

The design solution proposed is based on the values as presented in the table above at Filtration Plant inlet. All concentrations refer to max concentrations to be used for the system's design. **Any change in the actual inlet parameters will have impact on Process Design, Engineering Design, Cost and Performance.**

1.3. INFLUENT FLOW DATA

PARAMETER	VALUE	UNIT
Filtration Plant Feed Flow Rate	6000	M^3/DAY

1.4. OPERATIONAL BASIS

Hours Per Day of Operation	24
Days Per Year of Operation	350 days (Downtime for chemical cleaning and other regular annual maintenance work 15 days considered)

1.5. PROCESS COMPATIBILITY

1. The incoming raw wastewater shall not contain any substance incompatible with the MOC of the Equipment supplied & inhibitory substances that sufficiently affect the biological treatment stage so as to compromise the operation of any of the system.

2. DESIGN SYSTEM CHOSEN

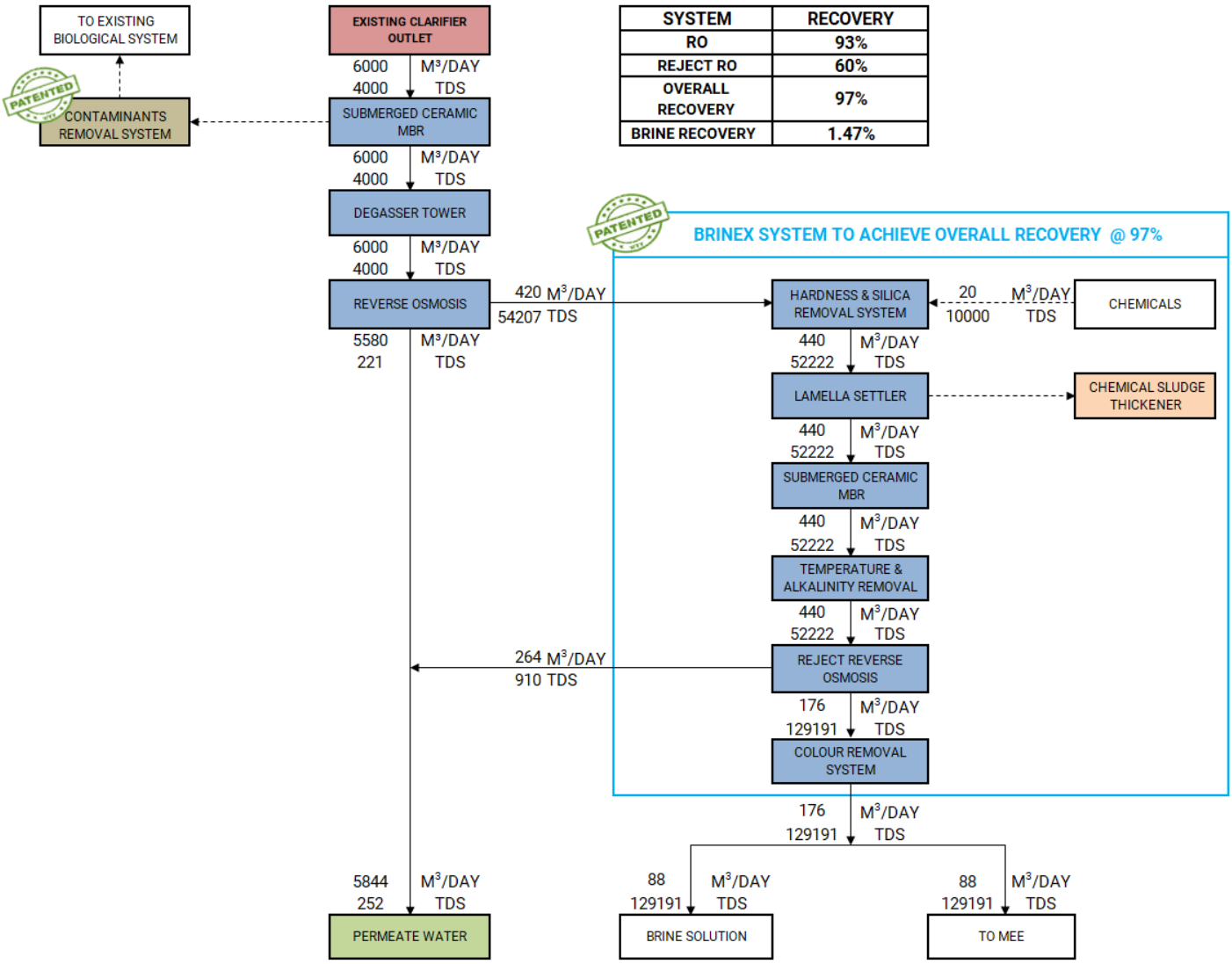
S.NO	PROCESS	CAPACITY (M^3/DAY)
1.	SUBMERGED CERAMIC MBR SYSTEM	6000
2.	CONTAMINANTS REMOVAL SYSTEM PATENTED	6000
3.	DEGASSER TOWER SYSTEM	6000
4.	REVERSE OSMOSIS SYSTEM – **(93% RECOVERY IN 4 STAGES)**	6000
	BRINEX SYSTEM PATENTED	
5.	HARDNESS & SILICA REMOVAL SYSTEM	440
6.	LAMELLA SETTLER	440
7.	SUBMERGED CERAMIC MBR SYSTEM	440
8.	TEMPERATURE & ALKALINITY REMOVAL SYSTEM	440

9.	REJECT REVERSE OSMOSIS SYSTEM **(60% RECOVERY IN 2 STAGES)**	440
10.	COLOR REMOVAL SYSTEM	176

***System capacities mentioned above are based on ZLD based Filtration plant influent flow Rate and the Actual capacities may vary.**

PROPOSED WATER BALANCE

ZLD BASED FILTRATION WITH BRINEX SYSTEM FOR 6000 M³/DAY @ 4000 TDS



SYSTEM	RECOVERY
RO	93%
REJECT RO	60%
OVERALL RECOVERY	97%
BRINE RECOVERY	1.47%

FILTRATION PLANT + BRINEX SYSTEM - STAGEWISE PARAMETERS				
PARAMETER	UNIT	FILTRATION PLANT INLET	FP + BRX @ 97% RECOVERY	
			PERMEATE	REJECT
pH	--	6.5 - 7.5	6 – 7	6 – 7
BOD	mg/L	< 30	BDL*	< 600
COD	mg/L	< 100	BDL*	< 1800
TDS	mg/L	4000	< 300	126000 – 131000#
TSS	mg/L	30	BDL	NA
TKN	mg/L	< 1	BDL*	< 20
P_4	mg/L	< 2	BDL*	< 80
PVA	mg/L	NIL	NIL	NIL
SULPHATE	mg/L	700	< 75	22000 – 25000
CHLORIDE	mg/L	600	< 75	19000 – 21000
REACTIVE SILICA	mg/L	< 15	< 1	< 300
HARDNESS	mg/L as $CaCO_3$	200	< 1	1700 – 1900
ALKALINITY	mg/L as $CaCO_3$	400	< 50	800 – 1100
TEMPERATURE	ºC	30 – 35	30 – 35	30 – 35
OIL & GREASE	mg/L	NIL	NIL	NIL
IRON	mg/L	< 0.01	BDL	NA

BDL – below detectable limit, NA – not applicable; Values are system generated which may vary with variation in inlet & on operational basis

2.1. EXPECTED TREATED WATER QUALITY

The following performance parameters are expected under standard operating conditions after equipment start-up based on the data and assumptions listed above and in Appendix, Warranties. In case of conflicting numbers, the ones listed under Appendix take precedence:

PARAMETER	TREATED WATER QUALITY	UNIT
Permeate Water TDS	200 – 300	mg/L

Notes:

- All influent quality parameters are based on daily average values of a minimum of four (4) Nos. of hourly composite samples collected at regular intervals over a day, with testing performed to applicable industry-approved standards.
- The above-mentioned recovery and water quality expectation is based on WTT supplying the system and scope Equipment as per the scope of supply table described in section 3 below.

3. Equipment Details for Proposed Systems

3.1. Submerged Ceramic MBR System

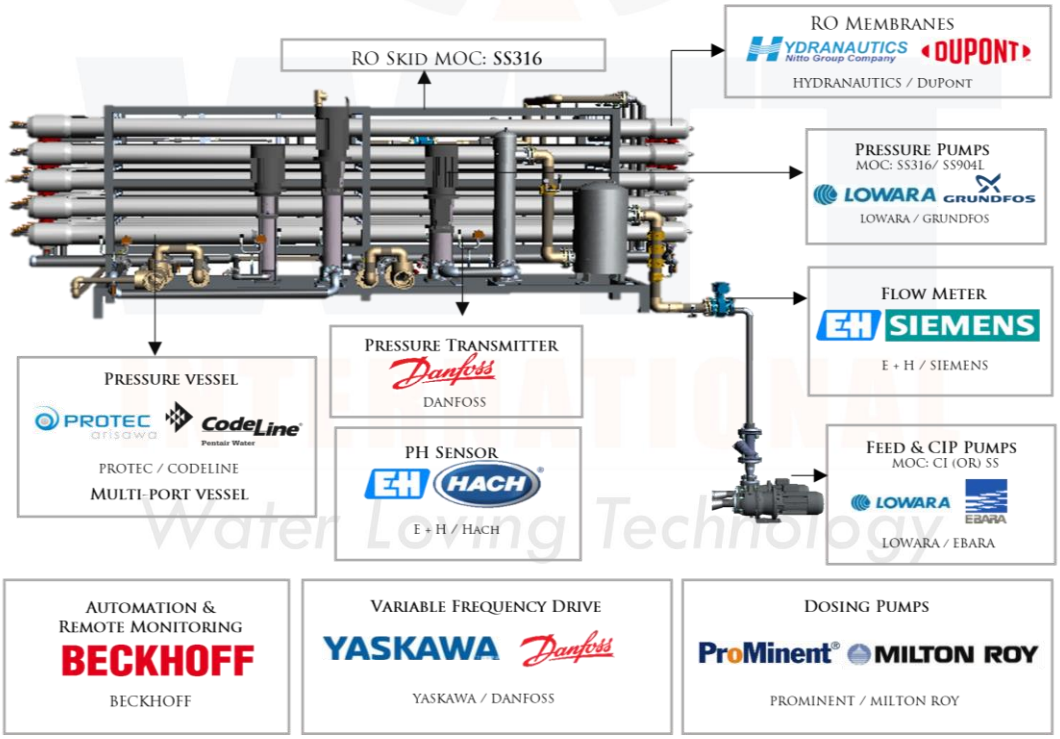
This system comprises of Eco-friendly & long-lasting Ceramic Membranes. Ceramic MBR system operates at low operational flux and easy to clean as the membrane is rigid and do not lose its form over a period of time. The system can be cleaned with Permeate & Chemically Enhanced Backwash technology and Air scouring to avoid sludge deposition. The sprinkler pumps ensures that there is no clogging over the membrane surface.

3.2. Degasser Tower System

Degasser tower eliminates the carbonaceous gases from the effluent water by means of a packed decarbonizing tower. The air blown from the bottom of the tower carries away CO_2 from wastewater to the top of the tower to reduce the dissolved gases and there by protect the RO membranes from scaling.

3.3. REVERSE OSMOSIS SYSTEM

Reverse Osmosis Process is the forced passage of water through a membrane against the natural osmotic pressure to accomplish separation of ions and water. It separates the dissolved salts from the feed water and reduces the TDS level, and this can be reused for industrial process. We can achieve lowest CAPEX and OPEX.



3.4. BRINE RECOVERY SYSTEM

Reject water from the Reverse Osmosis system can be further treated with an additional RO system along with BRINEX system to increase recovery. The BRINEX system helps to decrease the load of the evaporator there by reducing the high operating cost of Evaporator and also increase the permeate water quantity.

At **step 1**, Hardness & silica removal system is proposed where the reject water from the RO systems is raised to high pH, by addition of alkali for the conversion of Bicarbonates into Carbonates and reduction of Hardness causing magnesium, calcium ions into its hydroxide forms. This again treated with flocculants & coagulants for the precipitation of Hardness producing salts and silica removal. The effluent further Neutralized by the addition of acids. Hence, the effluent will have lesser amounts of Fouling and Scaling compounds.

At **step 2,** Submerged Ceramic MBR System comprises of Eco-friendly & long-lasting membranes. Submerged Ceramic MBR system operates at low operational flux and easy to clean as the membrane is rigid and do not lose its form over a period of time. The system can be cleaned with Permeate & Chemically Enhanced Backwash technology and Blower for Air scouring to avoid sludge deposition. The sprinkler pumps ensures that there is no clogging over the membrane surface. This system reduces suspends and provides clarified outlet water to be fed to RO membranes.

At **step 3**, Temperature & Alkalinity removal system designed where air and water brought into direct contact with each other in cross flow to reduce the inlet water temperature and carries away CO_2, dissolved gases from the effluent water for better performance in RO. This system proposed to protect the RO membranes from scaling and to increase the membrane life.

At **step 4**, The Reject RO increases the water recovery and the Reject stream from the Reject RO will have high TDS which will reduce the evaporator's load enabling lesser utility consumption. This increased water recovery reduces the feed to the evaporator. The High TDS solution can be reused as Brine solution in process plant.

At **step 5**, Colour removal system is proposed to reduce Colour and partial COD of the effluent for reusing it as a Brine solution and recovery of Colour less salt further by using MEE.

4. Equipment Scope of Supply– BY WTT

FILTRATION PLANT EQUIPMENT			
A	**SUBMERGED CERAMIC MBR SYSTEM**		**6000 M^3/DAY**
S.NO	**DESCRIPTION**	**SPECIFICATION**	**QUANTITY**
1.	SUBMERGED CERAMIC MBR SYSTEM	NO. OF MODULES	312 NOS.
		NO. OF TRAINS	3
		TOTAL FILTRATION AREA	2140 M^2
		MOC OF MODULE	CERAMIC
		PERMEATE PUMP	3W + 1SS
		PUMP MOC (CASING/IMPELLER)	CI/CI
		TYPE OF PUMP	SURFACE MOUNTED
		BACKWASH/CIP PUMP	1W + 1S
		PUMP MOC (CASING/IMPELLER)	CI/CI
		TYPE OF PUMP	SURFACE MOUNTED
		SLUDGE TRANSFER PUMP	1W + 1S
		PUMP MOC (CASING/IMPELLER)	CI/CI
		PUMP TYPE	SURFACE MOUNTED
		BLOWER	1W
		BLOWER TYPE	LOBE
		DOSING PUMP	2W
		DOSING PUMP DIAPHRAGM TYPE	PP
		TURBIDITY SENSOR	PERMEATE
		ELECTROMAGNETIC FLOW METER	PERMEATE
B	**CONTAMINANTS REMOVAL SYSTEM**		**600 M^3/DAY**
S.NO	**DESCRIPTION**	**SPECIFICATION**	**QUANTITY**
2.	CONTAMINANTS	LAMELLA SETTLER	1 LOT

	REMOVAL SYSTEM	LAMELLA PACKS MOC	PVC
		ANGLE OF INCLINATION FOR PACKS	55 – 60°
		LAMELLA SETTLER FRAME MOC	SS316
		DOSING PUMP	1W
		DOSING PUMP DIAPHRAGM MOC	PP
		SLUDGE PUMP	1W + 1S
		PUMP MOC (CASING/IMPELLER)	CI / CI
		PUMP TYPE	SURFACE MOUNTED
C	**DEGASSER TOWER**		**6000 M^3/DAY**
S.NO	**DESCRIPTION**	**SPECIFICATION**	**QUANTITY**
3.	DEGASSER TOWER	DEGASSER TOWER VESSEL	3
		TOWER MOC	FRP
		pH SENSOR	1
		DEGASSER TOWER MEDIA	3 LOT
		MEDIA MOC	PP
		DEGASSER TOWER BLOWER	3
		DOSING PUMP FOR DEGASSER TOWER	1W + 1SS
		DOSING PUMPS MOC	PP / TEFLON
D	**REVERSE OSMOSIS SYSTEM**		**6000 M^3/DAY**
S.NO	**DESCRIPTION**	**QUANTITY**	**QUANTITY**
4.	REVERSE OSMOSIS SKID	TOTAL NUMBER OF SKIDS	2
		SKID MOC	SS316
		NUMBER OF STAGES	4
		RO FEED PUMP	4W + 2S
		PUMP MOC (CASING/IMPELLER)	CI / CI
		PUMP TYPE	SURFACE MOUNTED
		PRE-FILTER	2 LOT

		PRE-FILTER HOUSING MOC	SS316
		DOSING PUMPS	6W + 1SS
		DOSING PUMPS MOC	PP
		DOSING TANKS	2
		DOSING TANK MOC	LDPE
		HIGH PRESSURE PUMP WITH INVERTER	4W
		HIGH PRESSURE PUMP MOC	SS316
		HIGH PRESSURE PUMP TYPE	VERTICAL/ MULTISTAGE
		BOOSTER PUMPS WITH INVERTER	6W
		BOOSTER PUMPS MOC	SS316
		BOOSTER PUMP TYPE	HORIZONTAL/VERTICAL-MULTI STAGE
		PRESSURE VESSELS	76 No's
		PRESSURE VESSELS MOC	FRP
		RO MEMBRANES	456 No's
		RO MEMBRANES MOC	COMPOSITE POLYAMIDE
		pH SENSOR	2
		PRESSURE INDICATOR	20
		PRESSURE TRANSMITTER	20
		ELECTROMAGNETIC FLOWMETER	(10 No's (RO FEED, STAGE-1,2,3,4 PERMEATE))
		CIP DOUBLE SKID WITH AUTOMATION (INCLUSIVE OF PUMPS WITH ITS FILTER, PH-SENSOR, FLOWMETER, PRESSURE TRANSMITTER, AUTO VALVES & DOSING PUMPS)	1 SET (COMMON FOR SKID 1&2)

BRINEX SYSTEM			
E	HARDNESS & SILICA REMOVAL SYSTEM		440 M^3/DAY
S.NO	DESCRIPTION	SPECIFICATION	QUANTITY
5.	HARDNESS & SILICA REMOVAL SYSTEM	PIPE FLOCCULATION SKID	1 LOT
		PIPE FLOCCULATION SKID MOC	PVC
		DOSING PUMP	4W
		DOSING PUMPS MOC	PP
		pH SENSOR	1
6.	LAMELLA SETTLER	LAMELLA SETTLER	1 LOT
		LAMELLA SETTLER FRAME MOC	SS316
		LAMELLA PACKS MOC	PVC
		SLUDGE PUMP	1W + 1S
		PUMP MOC (CASING/IMPELLER)	CI / CI
		PUMP TYPE	SURFACE MOUNTED
7.	SLUDGE THICKENER	PIPING FOR SLUDGE THICKENER	1
		MOC	SS316
G	SUBMERGED CERAMIC MBR SYSTEM		440 M^3/DAY
S.NO	DESCRIPTION	SPECIFICATION	QUANTITY
8.	SUBMERGED CERAMIC MBR SYSTEM	NO. OF MODULE	26 No's
		NO. OF TRAINS	1
		TOTAL FILTRATION AREA	178 M^2
		MOC OF MODULE	CERAMIC
		PERMEATE/BACKWASH/CIP PUMP	1W + 1S
		PUMP MOC (CASING/IMPELLER)	CI / CI
		TYPE OF PUMP	SURFACE MOUNTED
		DOSING PUMP	2W

		DOSING PUMP MOC	PP
		ELECTROMAGNETIC FLOW METER	PERMEATE
		TEMPERATURE & ALKALINITY REMOVAL SYSTEM	1 NO.
H	**REJECT REVERSE OSMOSIS SYSTEM**		**440 M^3/DAY**
S.NO	**DESCRIPTION**	**SPECIFICATION**	**QUANTITY**
9.	REJECT REVERSE OSMOSIS SKID	**TOTAL NUMBER OF SKIDS**	**1**
		SKID MOC	SS316
		NUMBER OF STAGES	2
		RRO FEED PUMP	1W + 1S
		RRO FEED PUMPS MOC	SS316
		PUMP TYPE	SURFACE MOUNTED
		PRE-FILTER	1
		PRE-FILTER HOUSING MOC	SS316
		DOSING PUMPS	2W + 1SS
		DOSING PUMPS MOC	PP
		HIGH PRESSURE PUMP WITH INVERTER	1W
		HIGH PRESSURE PUMP MOC	SS904L
		HIGH PRESSURE PUMP TYPE	HORIZONTAL/ MULTI STAGE
		BOOSTER PUMPS WITH INVERTER	1W
		BOOSTER PUMPS MOC	SS904L
		BOOSTER PUMP TYPE	HORIZONTAL/ MULTI STAGE
		PRESSURE VESSELS	9 No's
		PRESSURE VESSELS MOC	FRP
		RO MEMBRANES	54 No's
		RO MEMBRANES MOC	COMPOSITE POLYAMIDE
		pH SENSOR	1

		PRESSURE INDICATOR	6
		PRESSURE TRANSMITTER	6
		ELECTROMAGNETIC FLOWMETER	3 No's (RO FEED, STAGE-1 & 2 PERMEATE)
10.	COLOR REMOVAL SYSTEM	CIRCULATION PUMP	1W + 1S
		PUMP MOC (CASING/IMPELLER)	SS
		TYPE OF PUMP	SURFACE MOUNTED
		DOSING PUMP	1W + 1SS
		DOSING PUMPS MOC	PP

W – Working; S – Standby; SS – Store Standby

Specification and quantity details mentioned above are preliminary and may be revised to optimize the performance of the system

OTHER GENERAL ITEMS IN WTT's SCOPE:

- *Piping in SS, CPVC, UPVC and its associated items*
- *All other related valves and puddle flanges required*
- *Electrical control panel for proposed system with standard automation*
- *Power cable and control cable within the electrical panel*
- *Power cable and control cable from electrical panel to proposed equipment*
- *General layout & civil detailed drawing for proposed systems*

5. SCOPE OF SUPPLY & INSTALLATION- BY CLIENT

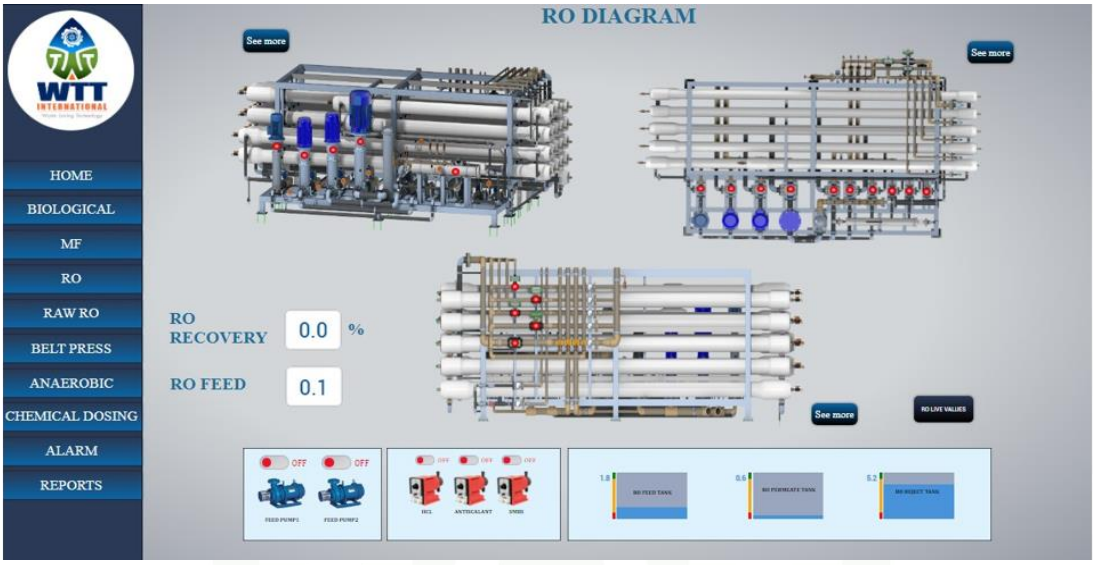
- Transfer of influent up to proposed Filtration Plant system.
- Bulk chemical storage tanks for proposed systems based on the available optimum level of chemical procurement
- Sludge Handling of proposed plant (Both Filtration plant & BRINEX System)
- Sludge disposal during plant commissioning and operation
- Backwash/CIP Tank (RCC) for Proposed MBR system
- Raw / Permeate water transfer for cleaning of filtration system
- Transfer of CIP cleaning water from Filtration systems
- Raw water supply for whole plant and treated water transfer (Both Permeate & Reject water)
- All underground piping & piping for electrical cable during civil works
- Incomer cable and its connections with WTT supplied electrical panel
- Two Separate & Distinct Earthing for all individual equipment & panel
- Instruments/Electronics Earthing & Safety/Protective Earthing shall be provided for individual Instruments, Motor/Pump & Other Electrical Equipment
- Input power to individual WTT supplied electrical panel
- All civil related works, civil structural drawings, underground piping & piping for electrical cabling during civil works.
- Installation of equipment supplied.
- All loading & unloading @ site
- All excavation works related with pipe and cables laying.
- Electrical cable & stabilized power supply until WTT panel's, Internal Individual Earthing as per requirement
- Utilities like Power, Steam, Hot water etc., if Required
- Chemical supply and storage/preparation tanks
- Separate panel, wiring with fans & lights for whole plant & its power supply
- Pre-Shipment Inspection requested by the local authorities.
- All license fees and / or custom duties.
- Provide Food, Accommodation and Local transport to the team of WTT.
- Installation consumables (such as welding electrodes, cutting disk and others).
- Supply and assembly of all drainage lines for rain water, sewers, etc.
- Any construction approval from local Authorities.

- Supply of any hoisting and/or laying equipment (cranes forklift, etc.) both truck mounted or not.
- Construction of site fencing/railing.
- Protection for all field instruments, local switches, motors, etc.,
- Safety handrails for all tanks, safety items like life buoy rings, eye washer, life-jackets, etc.
- Removal of all residual debris after construction and thorough cleaning of all tanks before plant start-up.
- General spares for supplied equipment.
- Any further item not described in our offer.

HIGHLIGHTS

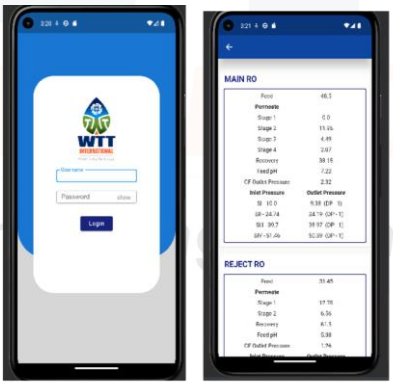
- *Our scope of supplies is of **SS316L pipes** for **Air line, RO & MBR pipe line,** materials like UPVC are used only in areas where SS316L is not compatible.*
- *Submerged CERAMIC MBR system with a **higher membrane life** and auto cleaning capabilities.*
- ***Rapid Filtration & Backwash Sequence** with automatic valves in pre-filtration.*
- *Specialized Degasser Tower proposed to **reduce the scaling of RO membranes** and in-turn achieves lesser number of cleaning cycles.*
- *This Degasser tower system enables to **increase the life of RO membranes.***
- ***Individual pressure transmitters for Inlet & Outlet** of each stage element to know exactly where the fouling has occurred and to initiate cleaning for that specific stage.*
- ***STATE OF ART 2 SKIDS of CIP system** for RO with full automation including dosing of chemicals with PH automation and flow with pressure control.*
- *Each stage of **RO permeate has individual EMFM** for better control and better automation to keep efficiency and life of RO.*
- *All part of the plant has **Level sensors & floats** to have complete data on volume of water in each tank.*
- *Electrical panel supplied by WTT with necessary cooling technologies and easy operative switch for Automatic & Manual operations.*

- ***PC controlled HMI panel*** *provided with complete automation and acknowledgement & warnings for cleaning necessities based on the operating conditions.*



Sample of our HMI screens and is a tentative reference

- *Making the plant In-hand through* ***SCADA*** *for Live Data monitoring using a smartphone (or) PC.*



REMOTE DATA MONITORING

6. Typical Equipment Vendor List

S.NO	EQUIPMENT	VENDORS
1.	PRESSURE PUMP	GRUNDFOS / LOWARA
2.	SURFACE MOUNTED PUMP	LOWARA / EBARA
3.	DOSING PUMP	PROMINENT / MILTON ROY
4.	MBR BLOWER	ROBUSCHI
5.	CERAMIC MBR MEMBRANES	OVIVO
6.	RO MEMBRANES	HYDRANAUTICS / DUPOINT
7.	PRESSURE VESSELS	CODELINE / PROTEC
8.	VARIABLE FREQUENCY DRIVE	YASKAWA / DANFOSS
9.	ELECTROMAGNETIC FLOWMETER	E+H / SIEMENS
10.	INLINE ANALYZERS	E+H / HACH
11.	LEVEL TRANSMITTER	E+H / SIEMENS
12.	LEVEL FLOAT	FAES
13.	PRESSURE TRANSMITTER	DANFOSS
14.	PRESSURE GAUGE	FORBES MARSHALL / MASS / WIKA
15.	AUTOMATION	BECKHOFF
16.	ELECTRICAL PANEL	RITTAL / HOFFMAN
17.	ELECTRICAL CABLE	LAPP / HUBER+SUHNER
18.	CONTAMINANT REMOVAL SYSTEM	**WTT INTERNATIONAL**
19.	LAMELLA SETTLER	
20.	TEMPERATURE & ALKALINITY REMOVAL SYSTEM	
21.	DEGASSER TOWER SYSTEM	
22.	RO SKID – 1 & 2	

The above-mentioned vendors are subjected to change based on the delivery terms for an equivalent vendor without compromising the quality and WTT reserves the right to change the vendor without any prior intimation.

7. Pricing Detail

PRICE FOR PROPOSED SYSTEMS		
S.NO	**SYSTEM/EQUIPMENT**	**PRICE IN USD**
1.	**SUBMERGED CERAMIC MBR SYSTEM** WITH CONTAMINANT REMOVAL SYSTEM	**$ 2,518,697**
2.	**DEGASSER TOWER SYSTEM**	**$ 118,281**
3.	REVERSE OSMOSIS SYSTEM (**93% RECOVERY** IN 4 STAGES WITH **SINGLE SKID**)	**$ 1,581,097**
4.	**BRINEX SYSTEM** PATENTED (WHICH INCLUDES **HARDNESS & SILICA REMOVAL SYSTEM, SUBMERGED CERAMIC MBR SYSTEM, TEMPERATURE & ALKALINITY REMOVAL SYSTEM, REJECT REVERSE OSMOSIS SYSTEM** (**60% RECOVERY** IN 2 STAGES WITH REJECT TDS UPTO **129191 mg/L** & **COLOR REMOVAL SYSTEM)**	**$ 942,234**
TOTAL PRICE FOR PROPOSED SYSTEMS		**$ 5,160,309**

All above mentioned prices given for the proposed systems with complete automation

S.NO	OPTIONAL PRICE FOR PROPOSED EQUIPMENT	PRICE IN USD
1.	**RO HIGH PRESSURE PUMP** - STORE STANDBY FOR ALL 4 STAGES	**$ 48,616**
2.	**RRO HIGH PRESSURE PUMP** - STORE STANDBY FOR ALL 2 STAGES	**$ 105,727**

- Pricing provided herein does not include any duties and taxes.
- Pricing provided herein is on **EX-Works** basis & does not include the freight/insurance charges
- Supervision of erection and commissioning of WTT supplied system shall be extra.

- Payment/commercial terms & conditions as per WTT's General Terms & Conditions of sale.
- Price valid for 15 days from the date of proposal due to market volatility

8. Happy Customers

AND 12 CETPs ACROSS THE COUNTRY

9. Channel Partners

10. Process Description Video

11. Commercial Terms And Conditions

11.1. TAXES

Taxes and Duties shall be extra as applicable. Pricing provided herein does not include any taxes of destination country.

If Tax Exemption is applicable, customer has to provide a copy to WTT of any applicable tax exemption certificates as issued by an approved taxation authority for the specific project location. Without an approved tax exemption certificate received by WTT all submitted invoices will include applicable tax. However we reserve the right to revise our price in case of tax exemption.

11.2. FREIGHT

All the Equipment and components foresee **Ex-works** delivery as per I.C.C. Incoterms-2010.

11.3. INVOICING AND PAYMENT TERMS

The pricing quoted in this proposal is based on the following payment terms, in principally agreed.

(1) 30% Advance Payment along with PO.

(2) 70% by irrevocable Letter of Credit payable at sight against Shipment documents.

Equipment shipment is contingent on receipt of earlier milestone payments.

11.4. EQUIPMENT SHIPMENT

Client and WTT will arrange a kick off meeting after contract acceptance to develop firm shipment schedule. This estimated delivery schedule assumes no more than 1 week for Client review of submittal drawings. Any delays in Client approvals or requested changes may result in additional charges and/or a delay to the schedule.

Material supply for the proposed plant in 6 - 7 months (approx.) is expected. The delivery schedule excludes the months of August & December, the delivery schedule is subject to review and adjustment. Partial shipments are allowed. In case of modifications to the volume of the supply after the emission of the order confirmation, WTT reserves the right to modify the delivery time and the agreed price.

11.5. FORCE MAJEURE

Please be advised that force majeure is applicable to the quotation provided. The unforeseen and uncontrollable events as defined in our terms and conditions, such as natural disasters, wars or pandemics & other events that prevent the parties from fulfilling their contractual commitments have occurred, impacting our ability to honor the terms outlined in this quotation. We will diligently review the situation and promptly communicate any necessary adjustments or potential delays in fulfilling the quoted services or products in compliance with the force majeure provisions outlined in our agreement.

11.6. PRICING NOTES

All prices will be quoted in **USD**. Any applicable tax of destination country is not included. Client will pay all applicable Local, State / Provincial, or Federal taxes & Duties. WTT may manufacture and source the Equipment and any part thereof globally in the country or countries of its choosing, provided that the Equipment complies with all of the requirements specified in this Agreement.

The Equipment delivery date, start date, and date of commencement of operations are to be negotiated. Title and risk of loss will transfer upon delivery in accordance with the INCOTERMS 2010. Commercial Terms and Conditions shall be in accordance with WTT's General Terms and Conditions of Sale as included in Appendix.

12. CLIENT SCOPE OF SUPPLY

12.1. SAFETY AND ENVIRONMENTAL

First aid, emergency medical response, eyewash & safety showers in the water treatment area. Chemical spill response, security & fire protection systems as per local codes. Environmental use and discharge permits for all chemicals at Client facility either listed in this document or proposed for use at a later date. Any special permits required for WTT's or Client employees to perform work related to the water treatment system at the facility. All site testing, including soil, ground and surface water, and air emissions, etc. Disposal of all solid & liquid waste from WTT's system including waste materials generated during construction, startup and operation.

Provide appropriate protection of the environment & local community, the health and safety of all workers and visitors at the site and the security of the facility. Provide safety related equipment & services such as site security, fire systems, lifting equipment and its operation, fall protection, adequate floor grating, ventilation, and safe access to equipment & electrical systems areas.

Equipment and trained support personnel for any confined space entry required during equipment installation/startup/commissioning/servicing. For permit-required confined space entry, a qualified rescue team on stand-by and available to respond within 4 minutes of an emergency.

Client will identify and inform WTT's personnel of any hazards present in the work place that could impact the delivery of WTT's scope of supply and agrees to work with WTT to remove, monitor, and control the hazards to a practical level. Client will provide training to WTT's personnel on all relevant & standard company operating procedures and practices for performing work on site.

12.2. JOBSITE AND INSTALLATION REVIEW

Review of WTT's supplied equipment drawings and specifications. All easements, licenses and permits required by governmental or regulatory authorities in Connection with the supply, erection and operation of the system.

Overall plant design, detail drawings of all termination points where WTT's equipment or Materials tie into equipment or materials supplied by others Stamping, signing or Sealing General drawings as per State, or local regulations or codes. All applicable civil design and works, including any building, site preparation, grading, excavations, foundations and trenches and accessories.

All electrical Labor and supplies leading up to jobsite including fittings, conduit, supports, cable trays, wire and hardware, air-conditioning of panels required to appropriately ground / earth the equipment as required for installation and ongoing operations.

All mechanical Labor and supplies leading up to jobsite including interconnecting piping, heat tracing (if required), fittings, conduit, pipe supports, and hardware as required for installation and ongoing operations. All instrumentation and automatic pneumatic valves

including but not limited to; air/sample line tubing, fittings, conduit, supports, isolating valves as required for installation and ongoing operations. Loading, unloading and transportation of equipment, materials required for WTT to perform duties outlined in WTT's Scope of Supply to the jobsite and/or warehouse.

Client will provide all access structures (scaffolding), mechanical lifting equipment (cranes, forklifts, scissor lifts, etc.), suitable site/shelter for placement of the proposed equipment, either inside appropriate housing, or outdoors.

Precaution: electrical equipment including the PLC may require air-conditioned rooms to prevent overheating of sensitive electronic equipment depending on climatic conditions. Bulk chemical storage and tanks, including secondary containment in accordance with local codes. Client will receive, off-load, log, and store all chemical and materials in accordance with Manufacturer's recommendation that are shipped. Compressed instrument air for pneumatic valves instruments and ejector system Equipment anchor bolts, Laboratory services, operating and maintenance personnel during equipment checkout, start-up and operation. Any on-site painting or touch-up painting of equipment supplied. Disposal of any Preservative.

For erection, WTT will ship the accessories such as pipes, bolts, nuts, screws, cables, valves etc., in excess quantities than required. This is in order to avoid shortages during the assembly due to damage or any other cause as it is difficult to procure the accessories locally. It is notified that all the shipped accessories during assembly and the excess accessories after commissioning are always remain the properties of WTT. WTT has the authority to retake the same at any period of time, even before and after commissioning. It is client's responsibility to preserve the above-mentioned items till WTT takes back the same.

12.3. START-UP AND COMMISSIONING

Installation & removal of temporary screens on all process lines Flushing and disinfection of all piping and tanks (including process equipment tanks) and verification of removal of all residual debris from construction. Alignments & required materials for rotating equipment MEG testing of all field motor power wiring (as required). Continuity checks for all electrical field wiring as per Installation Checklist, Hydro-testing of all field installed piping.

Supply raw materials, oil/lubricants chemicals and utilities during start-up and operation. Electrical & Mechanical support labor for commissioning activities, loading of membranes, stacks, modules, etc.

12.4. FACILITY MANAGEMENT

Client will provide such warehouse storage space and facilities, as are available at the site, and are reasonably appropriate to store parts, consumables, tools, etc. in accordance with manufacturer's recommendations. Such warehouse storage space will be a segregated area, secured and protected from adverse climate as may reasonably be required. The storage area shall be facilitated with 24 hours lock & key with security and Client will be responsible for risk of loss of WTT's parts while in storage at the site. Client will maintain WTT's parts stored at the site free & clear of any and all liens of Client and lenders, bondholders, contractors & other creditors of any nature.

Client will afford WTT's personnel free access and egress of the facility for all authorized work. Provide workshop facilities/area with roof and stabilized power supply, as is reasonably appropriate to carry out machining/fabrication works.

Provide adequate illumination and emergency lighting for all areas in which the WTT will be executing the scope of supply. Identify a Client project contact person to be available to WTT's personnel to address any issues related to WTT's execution of WTT's scope of work. Responsible for the equipment for movement of chemical drums, totes, and resins, as is reasonable. Provide all site utilities such as raw water, instrument quality air, potable water and power required for operation of the proposed equipment included in this scope of supply.

12.5. CONDITIONAL OFFERING

Client understands that this order confirmation has been issued based upon the information provided by Client and currently available to, WTT at the time of proposal issuance. Any changes or discrepancies in site conditions (including but not limited to system influent characteristics, changes in Environmental Health and Safety ("EH&S") conditions, and/or newly discovered EH&S concerns), Client financial standing, Client requirements, or any other relevant change, or discrepancy in, the factual basis upon which this proposal was created, may lead to changes in the offering, including but not limited to

changes in pricing, warranties, quoted specifications, or terms and conditions. WTT's offering in this proposal is conditioned upon a full WTT EH&S.

13. Appendix

13.1. APPENDIX A: CLARIFICATIONS

- Pump MOC has been considered as per WTT standard practice.
- The Equipment in WTT scope will be procured as per WTT Standard Vendor List.
- WTT has not envisaged third party equipment inspection before dispatch.
- The payments received once could not be processed again for refund.
- WTT has considered instruments as per WTT standard practice
- Painting specifications of WTT supplied equipment is as per standard manufacturer painting specifications.

13.2. APPENDIX B: ACCEPTANCE

To ensure accurate and prompt order entry, product delivery, billing and accounts receivables processing, please ensure your Purchase Order contains the complete Client and order details.

Please communicate your PO through:

E-mail	venkat@wttint.com
Postal Mail: (Indian Office)	WTT INTERNATIONAL PVT. LTD. No. 3, College cross road Avarangadu, Tiruppur, TamilNadu, INDIA – 641602

Bank Account details:

Beneficiary Name : WTT INTERNATIONAL PVT. LTD.,

Bank Name : INDIAN BANK LTD

Branch Name : TIRUPUR MSME BRANCH, TIRUPUR, TAMILNADU.

Account Number OD : 6475173650,

IFSC Code : IDIB000I043

13.3. APPENDIX C: WARRANTY

The mechanical warranty is only applicable to equipment supplied by WTT. The mechanical warranty period on all equipment supplied, unless otherwise noted, is twelve (12) months from the date of installation or fourteen (14) months from equipment shipment, whichever occurs first. WTT's obligation under this warranty is to the repair or replace, of any device or part thereof, which shall prove to have been manufacturing defects.

This warranty excludes the diffuser membranes, electrical items, defects, failures, damages or performance limitations caused in whole or in part by normal wear & tear, power failures, surges, fires, floods, snow, ice, lightning, excessive heat or cold, highly corrosive environments, accidents, actions of third parties, or other events outside of WTT's control. Warranty period for the entire equipment including replaced or repaired parts will be limited to the unexpired portion of the total warranty period. Bought out components are guaranteed only to the extent of guarantees given to us by our suppliers.

WTT assumes no liability for any damage to equipment caused by inadequate storage or handling as per manufacturer's recommendations in supplied technical literature, or by defective or sub-standard workmanship or materials provided by Client or any other third party responsible for handling, storing or installing the equipment. Client undertakes to give immediate notice to WTT if goods or performance appear defective and to provide WTT with reasonable time and opportunity to make inspections and tests. If WTT is not at fault, Client shall pay WTT the costs and expenses of the inspections and tests.

Goods shall not be returned to WTT without WTT's permission. WTT will provide Client with a "Return Goods Authorization" (RGA) number to use for returned goods. All return costs associated with shipping and labor are not included in the mechanical warranty. WTT warrants, subject to the provisions herein after set forth, that after stable operation of the WTT system has been attained and operators have acquired reasonable skills, the Equipment supplied for this project will be capable of producing the results set forth in stage wise parameter table, provided that:

The Equipment is operated and maintained at all times in accordance with the WTT Operations and Maintenance manual, The Equipment is operated within the mixed liquor characteristics defined in Influent quality table of this section, WTT has, until performance of its obligation herein is met, reasonable access to the Equipment and the operational data

relating thereto, Client furnishes adequate and competent operating, supervisory and maintenance staff, and necessary laboratory facilities with test equipment and personnel, Client utilizes the services of WTT until its performance obligations are met, Client supplies all necessary raw materials and services of a quantity and of a quality specified by WTT, An adequate and continuous power supply is available that will enable operation of all required equipment.